

Assign 4Q. 1.

$$\theta = \frac{T_s - T}{T_s - T_0}$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \left(1-D \text{ heat transfer via conduction} \right)$$

$$\eta = \frac{x}{b}, \quad \tau = \frac{t}{t^*}$$

$$t^* = ?$$

$$Fo = \tau = \frac{t \alpha}{b^2}$$

also known as Fourier Number

$$\left[\begin{aligned} \frac{\partial T}{\partial t} &= \alpha \frac{\partial^2 T}{\partial x^2} \\ \text{order of magnitude analysis} \\ \frac{(T_s - T_0)}{t^*} &\sim \alpha \frac{(T_s - T_0)}{b^2} \\ \Rightarrow t^* &= \frac{b^2}{\alpha} \end{aligned} \right.$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \xrightarrow{\text{Non-dimensionalisation}}$$

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial \eta^2}$$

B.C.

$$T = T_s \iff x = \pm b$$

$$\eta = \pm 1 \Rightarrow \theta = 0 \quad \forall t$$

I.C.

$$T = T_0 \iff t = 0$$

$$\tau = 0$$

$$\theta = \frac{T_s - T_0}{T_s - T_0} = 1$$

so let us try separation of variables technique:-

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial \eta^2}$$

$$\theta = f(\eta) g(\tau)$$

$$f \frac{dg}{d\tau} = g \frac{\partial^2 f}{\partial \eta^2}$$

$$\Rightarrow \frac{dg}{g d\tau} = \frac{1}{f} \frac{\partial^2 f}{\partial \eta^2} = -c^2$$

$$\frac{dg}{g dz} = -c^2 \Rightarrow g = A \exp(-c^2 z)$$

$$\frac{d^2 f}{d\eta^2} + c^2 f = 0 \Rightarrow \boxed{f = B \sin(c\eta) + D \cos(c\eta)} \quad \text{--- (a)}$$

B.C. $\eta = \pm 1$ $\theta = 0 \quad \forall t$

$f g = 0 \quad \forall t$

function of time, can take any value

hence $\eta = \pm 1$ $\underline{f = 0}$

~~applying B.C. for eqn (a),~~

~~or otherwise~~

we know temp profile is symmetric

$\Rightarrow f(\eta) = f(-\eta)$

$\Rightarrow f$ is even function

$f = D \cos(c\eta)$

apply B.C. for $\underline{f}$
 $\eta = \pm 1 \quad \underline{f = 0}$

$0 = D \cos(c)$

$\cos c = 0$

$\Rightarrow c_n = \left(n + \frac{1}{2}\right) \pi$

$n = 0, \pm 1, \pm 2, \dots, \pm \infty$

$\theta = \sum g_n f_n$

$= \sum \left(A_n \exp(-c_n^2 \tau) \right) D_n \cos(c_n \eta)$

$$\theta = \sum \underbrace{(A_n D_n)}_{\neq} \left(\exp \left(- \left(n + \frac{1}{2} \right)^2 \pi^2 \tau \right) \right) \cos \left(\left(n + \frac{1}{2} \right) \pi \eta \right)$$

$$= \sum_{n=0, \pm 1, \pm 2} P_n \left(\exp \left(- \left(n + \frac{1}{2} \right)^2 \pi^2 \tau \right) \right) \cos \left(\left(n + \frac{1}{2} \right) \pi \eta \right)$$

$$\theta = \sum_{n=0}^{\infty} Q_n \exp \left(- \left(n + \frac{1}{2} \right)^2 \pi^2 \tau \right) \cos \left(\left(n + \frac{1}{2} \right) \pi \eta \right)$$

using initial condition
 $\tau=0$ $\theta=1$

$$Q_n = \underbrace{P_n + P_{-(n+1)}}$$

$$1 = \sum_{n=0}^{\infty} Q_n \cos \left(\left(n + \frac{1}{2} \right) \pi \eta \right) \quad \text{--- (d)}$$

~~$\sum_{n=0}^{\infty} Q_n \cos \left(\left(n + \frac{1}{2} \right) \pi \eta \right)$~~

$$\frac{d^2 f}{d\eta^2} + c_n^2 f = 0$$

$$c_n = \left(n + \frac{1}{2}\right)\pi$$

$$\approx \eta = 0, \text{ from symmetry } \frac{df}{d\eta} = 0$$

$$f_n = \cos\left(\left(n + \frac{1}{2}\right)\pi\eta\right)$$

$$c_n = \left(n + \frac{1}{2}\right)\pi$$

$$\frac{d^2 f_n}{d\eta^2} + c_n^2 f_n = 0$$

$$\int_0^1 f_m \frac{d^2 f_n}{d\eta^2} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

$$f_m \frac{df_n}{d\eta} \Big|_0^1 - \int_0^1 \frac{df_m}{d\eta} \frac{df_n}{d\eta} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0$$

$$\left[f_m \frac{df_n}{d\eta} \right]_0^1 - \left[f_n \frac{df_m}{d\eta} \right]_0^1 - \int_0^1 \frac{df_m}{d\eta} \frac{df_n}{d\eta} d\eta + \int_0^1 c_n^2 f_n f_m d\eta = 0 \quad \text{--- (b)}$$

similarly starting with $\frac{d^2 f_m}{d\eta^2} + c_m^2 f_m = 0$
 we can prove that $c_m = \left(m + \frac{1}{2}\right)\pi$

$$- \int_0^1 \frac{df_n}{d\eta} \frac{df_m}{d\eta} d\eta + \int_0^1 c_m^2 f_n f_m d\eta = 0 \quad \text{--- (c)}$$

eqn (b) - (c)

$$(c_m^2 - c_n^2) \int_0^1 f_n f_m d\eta = 0$$

$$c_n^2 - c_m^2 = 0 \Rightarrow \underline{m = n}$$

if $m \neq n$, $\int_0^1 f_n f_m d\eta = 0 \Rightarrow f_n, f_m$
 orthogonal function

rewriting eqn. (d)

$$1 = \sum Q_n \cos\left(n + \frac{1}{2}\right)\pi\eta$$

$$\int_0^1 1 \times \cos\left(m + \frac{1}{2}\right)\pi\eta d\eta = \int_0^1 \sum Q_n \left(\cos\left(n + \frac{1}{2}\right)\pi\eta\right) \cos\left(m + \frac{1}{2}\right)\pi\eta d\eta$$

$$\int_0^1 \cos\left(m + \frac{1}{2}\right)\pi\eta d\eta = \int_0^1 Q_n \cos^2\left(n + \frac{1}{2}\right)\pi\eta d\eta$$

$$\frac{\sin(n+\frac{1}{2})\pi\eta}{(n+\frac{1}{2})\pi\eta} \Big|_0^1 = Q_n \int_0^1 \frac{1 + \cos((2n+1)\pi\eta)}{2} d\eta$$

$$\frac{(2)(-1)^n}{(n+\frac{1}{2})\pi} = Q_n \left(\eta \Big|_0^1 + \frac{\sin(2n+1)\pi\eta}{(2n+1)\pi} \Big|_0^1 \right)$$

$$\frac{(2)(-1)^n}{(n+\frac{1}{2})\pi} = Q_n \left(1 + \frac{\cancel{\sin} \cdot 0 - 0}{(2n+1)\pi} \right)$$

$$Q_n = \frac{2(-1)^n}{(n+\frac{1}{2})\pi}$$

$$\left[\theta = \frac{T_s - T}{T_s - T_0} = 2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+\frac{1}{2})\pi} \exp \left(-\left(n+\frac{1}{2}\right)^2 \pi^2 \left(\frac{d}{b^2}\right) \right) \right. \\ \left. * \cos \left(\left(n+\frac{1}{2}\right) \pi \frac{x}{b} \right) \right]$$